

Abhijit Sahu

abhijitsahu570@gmail.com | +91 7682066646 | Noida | LinkedIn: [Abhijit Sahu](#) | GitHub: [CODE-X-ABHIJIT](#)

PROFESSIONAL SUMMARY

Systems Engineer with hands-on experience designing, deploying and managing AWS and Kubernetes infrastructure for mission-critical banking applications. Skilled in Amazon EKS, Kubernetes, cloud networking, Linux, Docker, CI/CD, infrastructure automation, monitoring and cloud security. Experienced in building highly available cloud-native environments, automating infrastructure and deployments, optimizing resource utilization and troubleshooting distributed systems through root cause analysis (RCA).

EXPERIENCE

Systems Engineer Trainee

Noida, Uttar Pradesh

Shivalik Small Finance Bank

Dec 2025 – Present

- Design, provision and manage AWS and Kubernetes infrastructure for multi-environment banking platforms, implementing Amazon EKS workload isolation through taints, tolerations, node selectors and dedicated node pools to ensure controlled scheduling and resource separation. Provision and configure NGINX Ingress-based load balancers, including listeners, WAF policies and SSL/TLS certificates; configure server inbound security rules and network access controls, provision databases with secure connectivity from office premises and establish and manage IPsec tunnels across multi-cloud environments for secure network communication.
- Manage AWS infrastructure supporting internet and mobile banking platforms, maintaining high availability and platform stability during peak transaction workloads. Implement infrastructure configuration changes, auto-scaling strategies, security controls, firewalls and encryption standards to enable secure, resilient and seamless banking operations while supporting new application deployments.
- Engineer and maintain the cloud infrastructure for a critical banking customer onboarding platform on AWS and Kubernetes, ensuring high reliability, optimal resource utilization, security compliance and operational stability. Collaborate with software vendors to troubleshoot infrastructure, application and database issues, providing root cause analysis and technical guidance to accelerate issue resolution and maintain platform reliability.
- Oversee and maintain Enterprise Fraud Risk Management (EFRM) applications running on MicroK8s by troubleshooting pod and service failures, performing log analysis and resolving production incidents. Collaborate with cross-functional teams and vendors to ensure operational stability, regulatory compliance and uninterrupted banking services while supporting a real-time fraud detection platform powered by Kafka, Redis, Cassandra, Prometheus and Grafana.

PROJECTS

• RootIQ – Kubernetes Incident Detection & Auto-Remediation:

Tools: (Python, Kubernetes, PyPI Packaging, CLI Development, Rule Engine Architecture, Plugin Framework, YAML)

Developing **RootIQ**, an extensible Python-based CLI tool for automated Kubernetes inspection and diagnostics. Designed modular collectors and a rule-engine architecture to detect workload failures, configuration issues, security risks and resource misconfigurations across Pods, Deployments, Nodes, Ingresses and Network Policies. Implemented automatic rule discovery, plugin-based checks, severity-based incident reporting and standardized evidence and recommendations. Packaged as an installable CLI supporting **inspect**, **repair**, **validate** and **report** workflows, reducing manual Kubernetes troubleshooting effort through automated diagnostics, RCA, health reporting and remediation recommendations. ([REPOSITORY](#))

• FinOps Lab – Cost & Resource Optimization:

Tools: (Kubernetes, Docker, Prometheus, OpenCost, Metrics Server, HPA)

Built a local Kubernetes **FinOps Lab** using Kind with a 4-node cluster, deploying 5 containerized applications to simulate an overprovisioned environment. Integrated Prometheus, Metrics Server and OpenCost for performance and cost monitoring, reducing unnecessary CPU requests by approximately 79% through resource optimization. Implemented HPA for workload autoscaling and simulated node failures to validate Kubernetes self-healing and automatic pod rescheduling. ([REPOSITORY](#))

- **HawkEye – Automated Website Monitoring:**

Tools: (Python, GitHub Actions, FastCron, Prometheus, Grafana, Blackbox Exporter, Docker, SMTP (Gmail))
Developed **HawkEye**, a lightweight monitoring solution using Python, GitHub Actions, Prometheus, Grafana and Blackbox Exporter for automated health checks of multiple web applications. Implemented concurrent HTTP probing, real-time dashboards and email alerts for service failures. Automated monitoring workflows through FastCron and GitHub Actions, using GitHub Secrets for secure credentials and eliminating the need for dedicated infrastructure by leveraging free-tier services and open-source tools. ([REPOSITORY](#))

Certifications

- Cloud Computing and Internet of Things (IoT) — NPTEL
- Management Information Systems (MIS) — NPTEL
- AWS Certified CloudOps Engineer – Associate (In Progress)

SKILLS

Cloud Platforms: AWS (EC2, IAM, VPC, EKS, ECR, S3, CloudWatch, CloudTrail), **Container Orchestration:** Kubernetes, Helm, Docker, kubectl, **Infrastructure & Automation:** GitHub Actions, CI/CD, Terraform, **Monitoring & Observability:** Prometheus, Grafana, Kibana, Blackbox Exporter, **OS:** Linux, **Programming:** Java, Python, **Databases:** PostgreSQL, MS SQL, **Concepts:** Incident Management, Production Support, Disaster Recovery, Root Cause Analysis (RCA), High Availability, Cloud Security

Soft Skills: Problem-Solving, Adaptability, Critical Thinking

EDUCATION

Biju Patnaik University of Technology

Rourkela, Odisha

Master in Computer Application (MCA), Institute of Management and Information Technology, Cuttack

2023-2025

CGPA: 8.73

ACADEMIC ACHIEVEMENTS

- Engaged in several hackathons, showcasing innovative problem-solving and technical skills in competitive environments.
- Secured 1st Runner-Up position in University-Level Mathematics Olympiad, demonstrating strong analytical and problem-solving abilities.